W-L-312C(GSA-FSS)
February 25, 1985
SUPERSEDING
W-L-312B
October 8, 1975

FEDERAL SPECIFICATION

LIGHT, FLOOR OR DESK, FLUORESCENT, SWINGING ARM

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers adjustable, swinging-arm, fluorescent lights for use on drafting tables, desks, and floors.

1.2 Classification.

1.2.1 Types. Lights covered by this specification shall be of the following types, as specified (see 6.2):

Type I - Weighted base, desk model.
Type II - Weighted base, floor model.
Type III - Mounting clamp, desk or drafting table.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications

PPP-B-585 - Boxes, Wood Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.

Federal Standards

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies)
Fed. Std. No. 595 - Colors.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes the cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, D.C., Atlanta, Chicago, Kansas City, MO., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-T-704 - Treatment and Painting of Material.
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-P-1116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurements should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute:

ANSI C78.375 - Fluorescent Lamps - Guide for Electrical Measurements.

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ANSI C78.1 - Dimensional and Electrical Characteristics of Fluorescent Lamps.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 "P" Street, N.W. Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

Underwriters' Laboratories, Inc., Standards (UL):

UL-153 - Portable Electric Lamps
UL-486 - Wire Connectors
UL-542 - Lampholders, Starters, and Starter Holders for Fluorescent Lamps.
UL-817 - Cord Sets and Power Supply Cords.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, NY 11749; 207 East Ohio Street, Chicago, IL 60611; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 Preproduction model. When specified (see 6.2), a preproduction model shall be submitted for examination, test, and inspection in accordance with section 4, and such other tests as are necessary to determine compliance with the provisions of this specification and the contract. In all respects not specifically covered by this specification, the preproduction model shall represent the identical quality of materials, workmanship, and design which will be used in the fabrication of the production units. Any fabrication of equipment to contractual requirements without approval of the preproduction model shall be at the contractor's risk. The approved preproduction model shall be used as a guide to inspect production units and may be presented for acceptance as the last unit of the contract. Examination and testing, whenever practicable, will be conducted at the contractor's plant, but the

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Government reserves the right to perform any of the inspections or tests set forth in this specification at a laboratory selected by the Government (see 6.2).

3.2 Description. The lights shall consist of two fluorescent lamps, a reflector, lampholders, switch, swinging-arm assembly, base and electrical cord and shall conform to UL 153, UL 486, UL 542, UL 817, and when specified, MIL-STD-108 (see 6.2).

3.3 Standard commercial product. The lights shall, as a minimum, be in accordance with the requirements of this specification and except for acquisitions from the Federal Prison Industries or from the blind and severely handicapped as defined in Federal Acquisition Regulations subparts 8.6 and 8.7, shall be the manufacturer's standard commercial or commercial-type product.

A standard commercial product is a product such as an item, material, component, subsystem, or system (a) regularly used for other than Government purposes and (b) sold or traded to the general public in the course of normal business operations at prices based on established catalog or market prices (see Federal Acquisition Regulation 15.804-3 for explanation of terms).

A commercial-type product is a commercial product (a) modified or altered, without degrading the quality, appearance, or function of the commercial product, in compliance with Government requirements and as such is usually sold only to the Government and not through the normal catalog or retail outlets, or (b) identified, packaged or marked differently than the commercial product normally sold to the general public.

Additional or better features, which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the light being furnished.

3.4 Fire and casualty hazards. Each contractor shall submit to the contracting agency proof that the light to be supplied under this specification conforms to the applicable requirements of UL 153 regarding fire and casualty hazards. The label or listing mark of the Underwriters' Laboratories, Inc., will be accepted as evidence that the light conforms to this requirement.

3.4.1 In lieu of the label or listing mark, the contractor may submit independent proof satisfactory to the contracting agency that his light conforms to the applicable requirements of the published standards, including methods of tests, of UL 153.

3.4.2 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to assure acceptance of the material or appliance.

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3.5 Design.

3.5.1 Electrical Rating. Unless otherwise specified (see 6.2), the lights shall operate from a 115/125 volt, 60 hertz, alternating-current source. The lights shall have a power factor of not less than 85 percent when tested in accordance with paragraph 4.4.4.

3.5.2 Dielectric strength. Each light shall have a dielectric strength of 1000 volts AC plus twice the rated operating voltage when tested in accordance with paragraph 4.4.2. The lights shall comply after being tested as specified in paragraph 4.4.1.

3.5.3 Material. The material shall be as specified herein. Material not definitely specified shall be of the quality regularly used by the manufacturer in the product sold on the commercial market. The material shall be free from all defects and imperfections that might affect the serviceability and appearance of the finished product.

3.5.4 Devices. The lights shall be furnished with reflector, ballast, lampholders, lamps, switch, wiring, and all other devices necessary for operation.

3.6 Construction. The lights shall be constructed of metal. Bases, columns, and supports shall be fabricated from one or more of the following: iron, steel, brass, bronze, aluminum, aluminum alloy, zinc alloy, or magnesium alloy. White-metal castings (lead base) shall not be used in bases, columns, or supports. Each light shall consist of a metal base with one or more metal columns or other supports, on the upper end of which shall be mounted a reflector and reflector housing. The reflector may be integral with the housing or may be separate and enclosed by a housing. All joints shall be of the die cast type. The light shall be sufficiently sturdy to withstand, without damage, the application of any force or stress to which it will normally be subjected in service and shipment.

3.6.1 Exposed edges. Exposed edges of sheet metal or other metal parts shall be finished so that personal injury due to sharp edges will not occur under any normal circumstances.

3.6.2 Wiring. The only exposed wiring shall be from the base to the plug connector and shall enter the light through or at the base (see 3.6). All other wiring shall be entirely concealed.

3.7 Electrical Cord. Each light shall be provided with an electrical (NEMA 1-15P) cord having an exposed length of not less than 6 feet with a molded-on parallel blade, 3-wire (NEMA 5-15P) grounding-type attachment plug. The cord shall be 3 conductor, including equipment grounding requirements of UL 153. Unless otherwise specified (see 6.2), the color of the cord shall be an approximate match to the color of the light finish.

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3.7.1 Splices. All splices shall be made with insulated spring-type or one piece molded splice conductors in accordance with UL 486.

3.7.2 Polarity. When tested in accordance with 4.4.3, the polarity of the light shall meet the requirements of UL 153. The lights shall be wired without any component being attached or placed in the circuit between the switch and the ungrounded wire of the electrical cord.

3.8 Base.

3.8.1 Type I. The base of the type I light shall be of sufficient size and weight to support the light and accessories in any desired position without tilting. The base shall have a covering of spun or shaped steel not less than 0.018 inch thick or of zinc alloy pressure-cast diecasting not less than 0.065 inch thick. A felt pad, not less than 50 percent wool, or other suitable material not less than 3/64 inch thick, of a color to harmonize with the light finish, shall be cemented to the bottom of the base to prevent the light base from marring or damaging any surface on which the light is placed.

3.8.2 Type II. The weighted base of the type II light shall consist of iron or steel. The pedestal shall be 39 inches (nominal) in height, measured from the bottom of the base to the hinge at the top of the pedestal. Stabilizing weights shall be of cast metal construction held firmly within the enclosure in the base. Substitute material used for weights must have the approval of the contracting officer. The base area and weight shall prevent the light from overturning when the base is inclined 8 degrees from the horizontal in any direction.

3.8.3 Type III. The base of the type III light shall consist of either 2 clamps, spaced approximately 3 inches apart or a clamp not less than 3 inches wide secured by two steel studs spaced not less than 1 1/2 nor more than 3 inches apart. The clamps shall have a minimum attachment range of 0 to 2.0 inches and shall be constructed of forged or formed steel or comparable material. Clamps shall be removable to allow the light to be attached to any surface with not less than 2 screws or bolts. Mounting holes shall be provided and easily accessible. A felt pad, not less than 50 percent wool, or other suitable material not less than 3/64 inch thick, of a color to harmonize with the light finish, shall be cemented to the uppermost clamping surface to prevent marring or damaging any surface on which the light is placed. Use of specialized tools shall not be required for attachment to mounting surface.

3.9 Swinging-arm assembly. The swinging-arm assembly shall consist of 2 lightweight tubular arms, made from steel tubing not less than 0.0400 inch thick, joined by a vertically adjustable elbow joint. Either or both arms may be of the single or double-rod construction. The elbow joint shall maintain any setting from fully retracted to fully extended. The extension of the swinging-arm assembly shall be 24 or 34 inches minimum, as specified,(see 6.2). Dimensions shall be determined with the swinging-arm assembly extended such that the reflector housing is in the

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down position and parallel to the mounting surface. Measurement shall be taken from the point where the assembly arm containing the electrical wiring is attached to the base ballast housing assembly directly to the point where the swinging-arm assembly is attached to the reflector housing. The bottom of the reflector housing shall be 12 inches above the surface on which the light is positioned during the measurement. All connecting joints shall allow the light to retain any selected position without sagging or other distortions. The swinging-arm and reflector housing assembly shall be counterbalanced by spring or other devices, such that the assembly shall maintain any selected position above the base and through at least 180 degrees horizontally around the base. For type I, the performance is the same except only through 90 degrees horizontally around the base. Any oscillation or vibration shall cease in less than 15 seconds(unit of time) after adjustment. The die cast joints shall retain their settings by means of balance mechanisms or compression-spring loaded friction joints; thumbscrews, wing nuts, or other methods of locking the adjustments shall not be employed.

3.10 Reflector housing assembly. The reflector housing assembly shall contain the reflector, switch, lamp and lampholders. The lamps shall be two 15-watt, 18-inch, T-8, cool white fluorescent lamps conforming to ANSI-C78.1.

3.10.1 Housing. The housing shall conform to UL 153 and shall be attached to the upper end of the extension arm by means of a die cast universal socket. The socket shall permit the reflector housing to be rotated a minimum of 180 degrees in each of 3 separate perpendicular planes and shall maintain the housing in the above position by means of compression-spring loaded or factory set friction joints; thumbscrews, wing nuts, or other means of locking the light in position shall not be employed. After being tested in accordance with paragraph 4.4.1, the light shall retain any selected position without oscillating over 15 seconds (unit of time) and shall not exhibit any distortion, abrasion of electrical insulation or any condition which will affect the safe use of the light. The housing shall afford free insertion and removal of both fluorescent lamps. When the lamps are in the burning position, no portion of the lamps shall project beyond the reflector rim.

3.10.2 Reflector. The reflector shall be made of sheet or other rolled metal bonderized and coated with a smooth, high gloss, white enamel. The reflector shall be securely attached to the reflector housing in a manner to increase the rigidity of the assembly and prevent movement of the reflector.

3.10.3 Switch. The switch shall be of the 2 button manual type and shall conform to UL 153 for use in this application. The switch shall be securely mounted in the reflector housing by screw(s) or locknut(s).

3.10.4 Lampholders. Lampholders shall permit easy insertion and removal of lamps and shall not soften or distort at or below 250 degrees F. Contacts shall be held securely in place and shall be insulated from

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one another. Lampholders shall conform to the applicable requirements of U.L. 542.

3.11 Ballast. The ballast furnished with the light shall be suitable for operation on the voltage and amperage specified and shall be in accordance with UL 153. A capacitor, when used for power factor correction, shall not be connected between the electrical power wire and the neutral ground wire in any manner which causes any type of electrical discharge across the prongs of the electrical cord.

3.12 Performance. The light shall retain any position setting without movement or distortion when tested as specified in paragraph 4.4.1.

3.13 Interchangeability. When specified (see 6.2), all parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable and the item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.14 Finish. All exposed metal parts shall be cleaned, free of dirt, oil, grease and moisture, treated and painted in accordance with UL 153 or when specified in accordance with MIL-T-704 (see 6.2). All surfaces except the reflector shall be smooth baked enamel and the finish color, unless otherwise specified (see 6.2), shall be an approximate match to color number 20122 of Fed. Std. No. 595. All enamel surfaces shall be free from tears, star marks, crazing, blisters, hairlines, chipping, pinholes, and other irregularities or defects.

3.15 Nameplate. Unless otherwise specified in the Invitation for Bids (see 6.2), an identification plate permanently and legibly providing the information required by UL 153 shall be securely attached to the light in an inconspicuous location.

3.16 Workmanship. The lights shall be free from defects which detract from their appearance or impair their serviceability. Electrical connections shall be secure, metal joints and seams shall be tight. The entire reflector and light assembly shall be firmly seated at all points and locked with bolts, nuts, and washers wherever necessary. Chipping, discoloration or abrasion of the surface finish shall not be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for examination and test. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all examination and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities, suitable to the Government, for the performance of the examination and test requirements specified herein. The Government reserves the right to perform any examinations or tests set forth in the specification where such

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examinations or tests are deemed necessary to assure that the supplies and services conform to the prescribed requirements.

4.1.1 Acceptance. Desk and floor lights furnished under this specification shall conform to all the requirements of this specification. Except as otherwise specified herein, for acceptance purposes, evidence of compliance with UL shall be the UL label or listing mark of the Underwriters Laboratories, Inc., or other Government approved nationally recognized testing organization properly equipped and qualified for testing to the appropriate Underwriters Laboratories, Inc., standard. Nonconformance with any requirements of this specification shall be cause for rejection.

4.2 Sampling. The unit of product for sampling shall be one completely assembled light.

4.2.1 For examination and tests. Sampling for examination and tests shall be conducted in accordance with MIL-STD-105 and as indicated in table I.

TABLE I. Sampling for examination and tests.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-1	2.5
For Tests	S-2	2.5

4.3 Examination. The light shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III. Examination shall not be limited to the classified possible defects listed below.

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TABLE III. Classification of defects.

Categories	Defects
MAJOR 100	
101	Component part missing or not as specified.
102	Any adjustable part fails to move easily.
103	Electrical rating other than required.
104	Swinging-arm assembly will not maintain selected position.
105	Reflector housing will not maintain selected position.
106	Unfinished or corroded areas.
107	Length of swing arm not as specified.
108	Construction of the light not as specified.
109	Material not as specified.
110	Does not maintain position without tilting. (Type I & II)
111	Workmanship.
MINOR 200	
201	Identification marking not as specified.
202	Assembled light exhibits dirt, oil, grease, chips, or foreign matter.

4.4 Tests.

4.4.1 Performance. The light shall be initially fully retracted with the reflector housing pointing down and as close to the inner arm as the design permits. The arm shall then be fully extended and the reflector housing rotated in each direction according to paragraph 3.10.1 and finally returned to the original configuration. The light shall be examined after completion of 150 cycles of movement for distortion and for failure to retain any position setting. The wiring shall be examined for any sign of abrasion and the dielectric strength test performed as specified in 4.4.2.

4.4.2 Dielectric strength. The light shall be tested for dielectric strength as specified in UL 153.

4.4.3 Polarity. A continuity check shall be made to determine if the switch is wired directly to the ungrounded conductor terminal on the plug as specified in UL 153.

4.4.4 Power factor. Power factor is the ratio of true power (watts) to the apparent power (volts x amperes) and shall be determined as specified in ANSI C78.375.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with section 5 of this specification. Defects shall be classified in accordance with table I. For examination of preparation for delivery, the sample unit

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shall be one shipping container packaged, packed, marked, and ready for shipment. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects.

300	MARKINGS	400	MATERIALS
301	Omitted	401	Missing components
302	Illegible	402	Damaged components
303	Incorrect		
304	Improper size		
305	Location	500	WORKMANSHIP
306	Sequence	501	Inadequate packing
307	Method of application	502	Inadequate stapling
		503	Bulging or distortion
		504	Loose straping
		505	Incomplete closure

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each light shall be packaged in accordance with method I02 of MIL-P-1116, Die-cut blocking and bracing shall be provided to prevent movement and damage.

5.1.2 Level B. Each light shall be packaged in a close-fitting box conforming to PPP-B-636, class-domestic. Die-cut blocking and bracing shall be provided to prevent movement and damage. The box shall be closed in accordance with the appendix to PPP-B-636.

5.1.3 Level C. The lights shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Unit packages of lights of like description shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, weather resistant; or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall fiberboard box and the wood

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boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Unit packages, in quantities as specified (see 6.2), of lights of like description shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C. Lights packaged, as specified in 5.1, shall be packed in a close fitting grade 275 (minimum) corrugated fiberboard box conforming to section 6(a), Rule 41 of the Uniform Freight Classification, and Rule 222 of the National Motor Freight Classification. Closure of the boxes shall be in accordance with Section 7 of Rule 41 of the Uniform Freight Classification. The light shall be protected within the box, using die-cut or folded synthetic cushioning material as necessary to prevent movement and damage during shipment, handling, or storage.

5.3 Markings. In addition to marking required by the procurement documents, the shipping container shall be marked in accordance with Fed. Std. 123 for civilian agencies and MIL-STD-129 for Military agencies as applicable (see 6.2).

6. NOTES

6.1 Intended use. The lights are intended for use on desks, tables, drawing boards, benches, etcetera, to shield the light from the workers' eyes and direct maximum intensity on the work area.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in their procurement documents:

(a) Title, number, and date of this specification.
(b) Type of light required (see 1.2).
(c) Preproduction model, if required and laboratory where preproduction inspection shall be conducted (see 3.1).
(d) Compliance with MIL-STD-108 if specified (see 3.2).
(e) Electrical rating if other than specified (see 3.5.1).
(f) Cord color if other than specified (see 3.7).
(g) Length of swinging-arm assembly (see 3.9).
(h) Compliance with MIL-D-1000 when specified (see 3.13).
(i) Compliance with MIL-T-704 when specified (see 3.14).
(j) Finish color if other than specified (see 3.14).
(k) Nameplate information, if specified, shall be in accordance with MIL-STD-130 (see 3.15) DoD only.
(l) Responsibility for inspection if other than specified (see 4.1).
(m) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(n) Marking if other than specified (see 5.3).

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